Bùi Huy Hoàng

GAME DEVELOPER (UNITY / C#)



ABOUT ME

Dedicated Game Developer specializing in Unity (C#) with strong software architecture rigor. Proven experience in implementing combat loops, Finite State Machines (FSM), generic object pooling with 0 B GC allocations, and custom shader batching with MaterialPropertyBlock. Seeking to contribute to an agile game studio and accelerate production iterations with AI workflows.

CONTACT

- hoangmon.pt@gmail.com
- github.com/hoangmonpt-cell
- gitlab.com/hoangmonpt-cell
- personal-portfolio.web
- Vietnam (On-site / Remote)

SKILLS

Game Engine & Languages:
- Unity 6 LTS, C#, .NET Standard, URP, Component-Based.

Gameplay & State Systems:
- FSM State Machines, Combat Logic, Tower Defense, Touch Physics.

Runtime Optimization:
- Generic Object Pooling (0 B GC Alloc), MaterialPropertyBlock batching.

Data & UI:
- ScriptableObject Architecture, UI Toolkit, UGUI, BigNumber math.

Web, CI/CD & AI:
- TypeScript, React 19, Three.js, GitHub Actions CI/CD, Vitest, AI Agents.

EDUCATION

Game Development & Software Engineering — 2024 – 2026

FPT Polytechnic

- Specialization: Unity Engine, C# Game Systems Architecture, Data Structures & Algorithms.
- Game Testing & QA (GAM203), 3D Game Development Workshops (GAM301/GAM302).

FEATURED GAME PROJECTS

Fortress Idle: Last Stand — 2026

Game Engineer & System Architecture | GitHub Repository

- **Zero-Alloc Combat Loop (0 B/frame GC Alloc):** Architected generic object pooling (`SimplePool<T>`) for projectiles, creeps, and floating damage numbers, completely eliminating GC-related frame stutters during combat.
- **Draw Call Batching via MaterialPropertyBlock:** Implemented hit-flash feedback without duplicating materials, preserving URP batching and GPU instancing.
- **Spatial Entity Registry:** Reduced target lookups from O(N) to O(1) via cached spatial arrays in `EnemyManager`, avoiding costly physics overlap scene scans.
- **Piecewise BigNumber Math Formula:** Designed mathematical scaling curve with 15 Engineering notation suffixes, supporting progression up to 10^45 without numeric overflow.

Ottershi (Unity 2D Mobile Port) — 2026

Gameplay & Systems Developer | GitLab Repository

- **Decoupled Manager - Controller Architecture:** Designed modular architecture separating overall game coordination (`GameManager`), conveyor belt mechanics, and sushi state machines (`SushiController`).
- **Touch Physics & Elliptic Trajectory:** Implemented elliptic trajectory math for conveyor motion and responsive touch-swipe throw velocity physics.
- **Fever Mode & Data-Driven Balance:** Built combo multiplier mechanics and configured game balance via decoupled `ScriptableObject` data containers.

3D Stylized Developer Room & Engine — 2026

Full-Stack 3D Web Engineer | GitHub Repository

- **3D Room Diorama & Raycasting:** Engineered interactive 3D room with camera raycasting and seamless switching between 3D Room and 2D Sci-Fi HUD.
- **Dual-API Synchronization:** Built resilient repository sync connecting GitHub REST API and GitLab Projects API with dual-tier fallback caching and GitHub Actions CI/CD.